

Auburn Forecast Temperature Report

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Summary

This report asks: **How do forecast temperatures change across the available Auburn forecast periods?** It uses the committed National Weather Service response and the project's existing validation and processing code. The answer is a concise description of the forecast temperature range, followed by the period-level evidence.

The committed forecast contains **14 periods**. Temperatures range from **72°F** during **Tonight** to **92°F** during **Today**. The period average is **80.7°F**.

Data and Parameters

The source is the National Weather Service points and gridpoint forecast APIs. The location is latitude 32.6099 and longitude -85.4808, representing Auburn, Alabama. The raw response is stored in `data/raw/forecast.json`. The report includes every period in that response without additional filtering. Temperatures use the unit supplied by the API. The report was rendered on July 30, 2026.

The project function `load_forecast_records()` reads the committed raw response, then calls the same Pydantic validation and transformation behavior used by the command-line interface.

Forecast Table

Table 1: Forecast periods and temperatures

#	Period	Starts	Temperature
1	Today	Jul 30, 07 AM	92°F
2	Tonight	Jul 30, 06 PM	72°F
3	Friday	Jul 31, 06 AM	92°F
4	Friday Night	Jul 31, 06 PM	75°F
5	Saturday	Aug 01, 06 AM	86°F
6	Saturday Night	Aug 01, 06 PM	73°F
7	Sunday	Aug 02, 06 AM	85°F
8	Sunday Night	Aug 02, 06 PM	73°F
9	Monday	Aug 03, 06 AM	86°F
10	Monday Night	Aug 03, 06 PM	73°F
11	Tuesday	Aug 04, 06 AM	87°F
12	Tuesday Night	Aug 04, 06 PM	74°F
13	Wednesday	Aug 05, 06 AM	88°F
14	Wednesday Night	Aug 05, 06 PM	74°F

Temperature Pattern

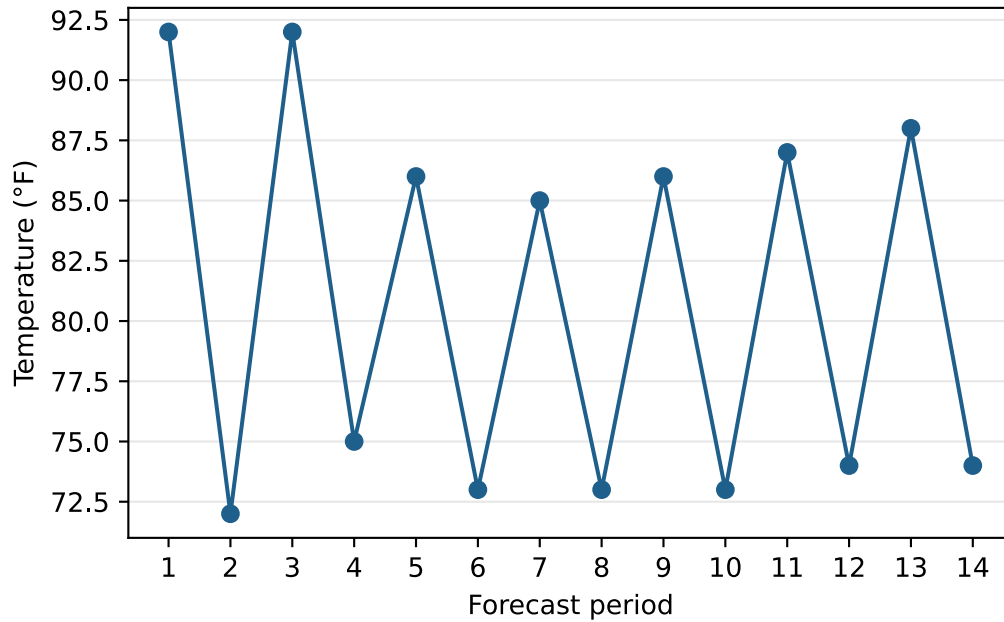


Figure 1: Forecast temperature by period in chronological order

Interpretation

The figure shows the expected alternating pattern between warmer daytime periods and cooler nighttime periods. The difference between the highest and lowest forecast values is **20°F**. These values are forecasts rather than observations, so they describe the service's current prediction and should not be interpreted as measured temperatures or forecast accuracy.

Rebuild

After restoring the environment with `uv sync` and installing Quarto, run this command from the repository root:

```
quarto render reports/auburn-forecast-report.qmd
```